# Deep Document Intelligence in Enterprise Workflows

A Comparative Analysis of Multimodal Layout Models and Heuristic Routing

Abstract?Enterprise document intelligence systems must process diverse layout typologies without manual intervention. We evaluate deterministic reading order routing across complex multi-column documents.

## 1. INTRODUCTION

Modern enterprise pipelines routinely encounter complex multi-column service brochures, technical manuals, and legal contracts. A naive top-to-bottom reading order merges text across column gutters, corrupting semantic context for downstream retrieval-augmented generation.

Deterministic horizontal band segmentation clusters words by vertical separation before applying column sorting. This prevents full-width section titles from being partitioned into column fragments.

## 2. FOOTNOTE HANDLING

Footnotes located at the bottom of the page [1] must not interleave with the terminal paragraphs of either column.

## 3. EXPERIMENTAL VALIDATION

We evaluate precision and recall across synthetic and real-world benchmarks. Multi-column documents demonstrate robust column boundary detection when gap thresholds adapt dynamically to local line heights.

## 4. CONCLUSION

Heuristic routing provides deterministic guarantees and sub-second processing latencies, making it an ideal primary ingestion engine prior to selective neural invocation.

[1] Document processing pipelines require deterministic reading order.